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(AUTONOMOUS)

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FMCW Radar-Based Mobile Surveillance Rover with Embedded TinyML Motion Classification

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Objective

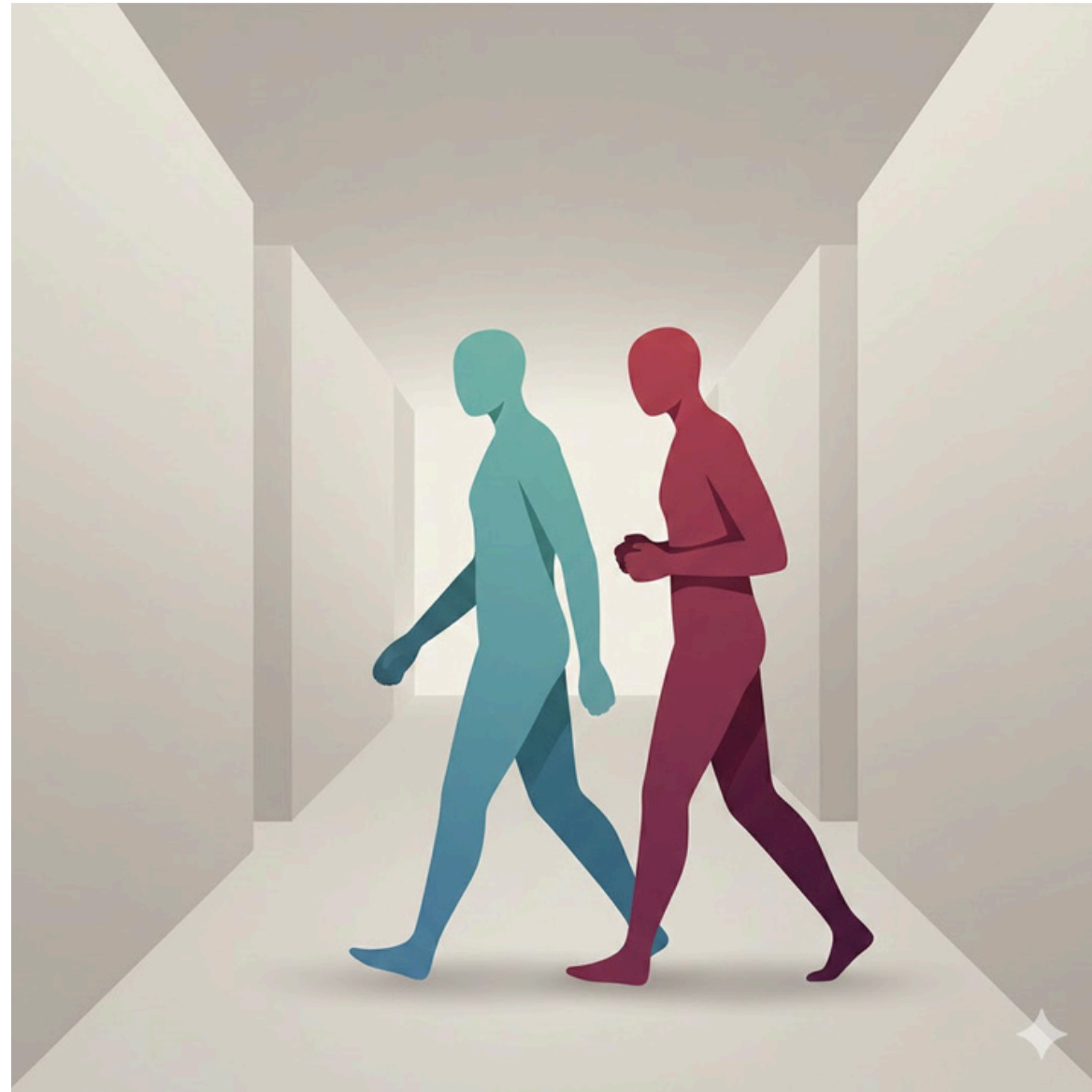


Fig 1: Visual Indicators of High Stress Vs Neutral Intent

- To develop a mobile FMCW radar-based rover for real-time detection and tracking of human targets.
- To implement TinyML-based motion classification of human behaviours such as static, creeping, walking, running, and pacing.
- To provide real-time visualization and remote teleoperation for practical indoor surveillance applications.

Introduction

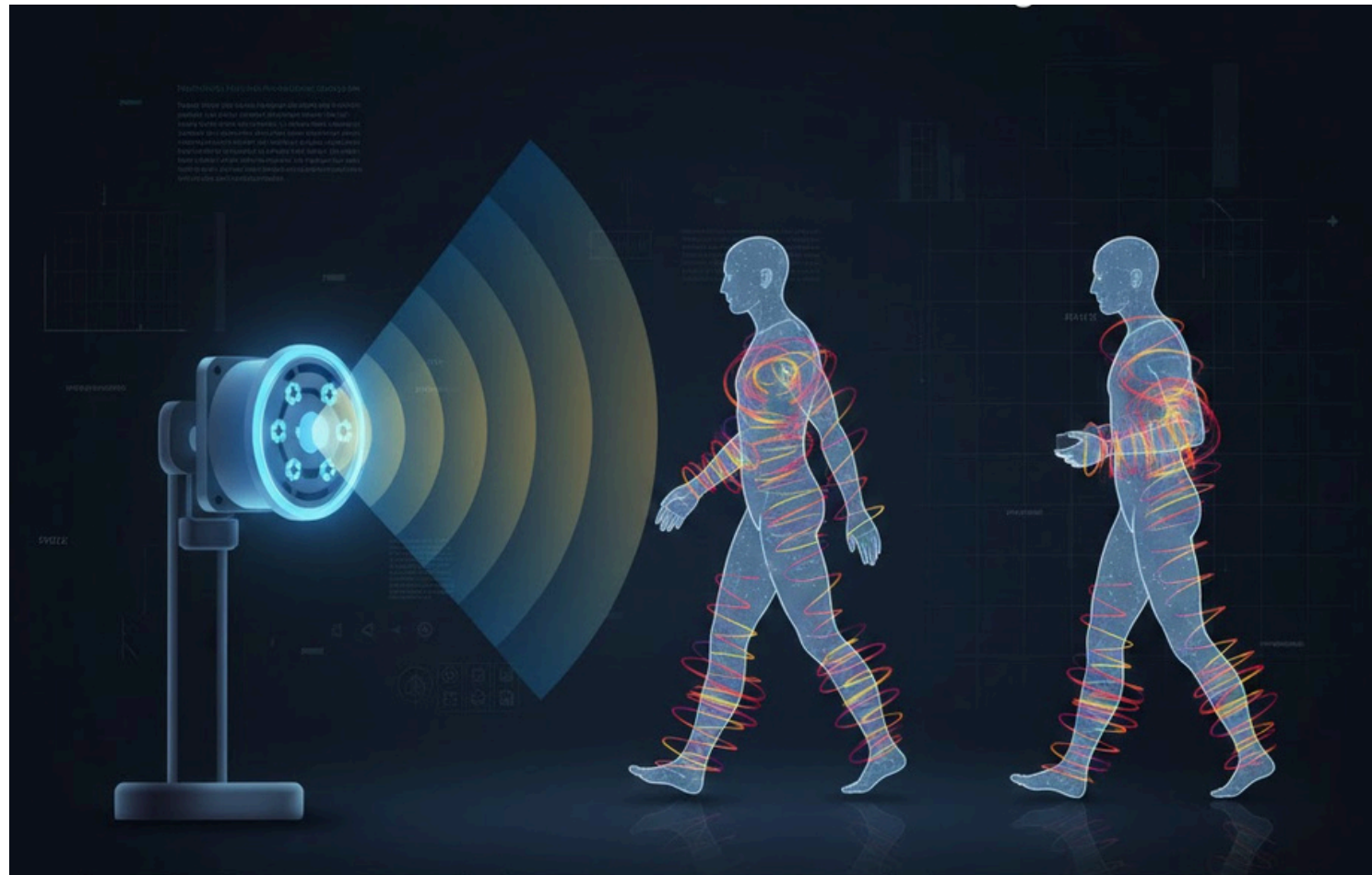


Fig 2 : Radar Intelligence Sensing

- Modern surveillance requires reliable detection that operates independently of lighting conditions and visual obstructions.
- Millimeter-wave radar provides privacy-preserving, real-time detection of human presence even in low-visibility environments.
- This project develops a compact standalone radar-based rover capable of multi-target tracking, motion-based behavior analysis, and real-time threat or alert evaluation for surveillance.

Literature Survey

Reference No.	Article Details	Journal Name	Year of publication	Observations
[1]	Multisensor Low-Cost System for Real Time Human Detection and Remote Respiration Monitoring	IEEE	2019	A pretrained neural network (MobileNet) is used to process camera frames and detect human presence in real-time.
[2]	A High-Computational Efficiency Human Detection and Flow Estimation Method Based on TOF Measurements	Sensors (MDPI)	2019	The proposed algorithm mainly consists of head detection based on local pooling and searching, classification refinement based on human morphological features.
[3]	Real-Time Human Detection and Gesture Recognition for On-Board UAV Rescue (Liu, C. and Szirányi, T.)	Sensors (MDPI)	2021	UAV rescue system using hierarchical gesture recognition (DNN/CNN). Key: dynamic "Attention" (set) and "Cancel" (reset) gestures.

Literature Survey

Reference No.	Article Details	Journal Name	Year of publication	Observations
[4]	A Human-Detection Method Based on YOLOv5 and Transfer Learning Using Thermal Image Data from UAV Perspective for Surveillance System (Mantau, A. J. et al.)	Drones (MDPI)	2022	Optimizes YOLOv5s with Transfer Learning for small-object human detection using RGBT (Thermal + RGB) images from a UAV. Achieved up to 88.8% AP on TIR images.
[5]	Using Deep Learning with Thermal Imaging for Human Detection in Heavy Smoke Scenarios	Sensors (MDPI)	2022	A thermal imaging camera (TIC) with a deep learning model is used as an intelligent human detection approach during emergency evacuations.
[6]	MILLIMETER WAVE RADAR-BASED HUMAN ACTIVITY RECOGNITION FOR HEALTHCARE MONITORING ROBOT (Gu, Z. et al.)	arXiv	2024	Proposes RobHAR for real-time, continuous HAR on a movable robot. Uses lightweight LPN-BiLiLSTM model with HMM-CTC for robust transition optimization.

Literature Survey

Reference No.	Article Details	Journal Name	Year of publication	Observations
[7]	Radar-Based Human Activity Recognition: A Study on Cross-Environment Robustness (El Hail, R. et al.)	Electronics (MDPI)	2025	Solves HAR domain shift in unseen environments. Combines time-frequency warping (DA) and DANN (UDA) to reduce performance drop from 29.6% to 13.2%.
[8]	Through-the-Wall Radar Human Activity Recognition WITHOUT Using Neural Networks (Gao, W.)	arXiv	2025	Non-NN TWR HAR. Segments micro-Doppler using ACM. Classifies using Mapper algorithm (topological similarity). Showed comparable robustness to noise as NNs.

BLOCK DIAGRAM

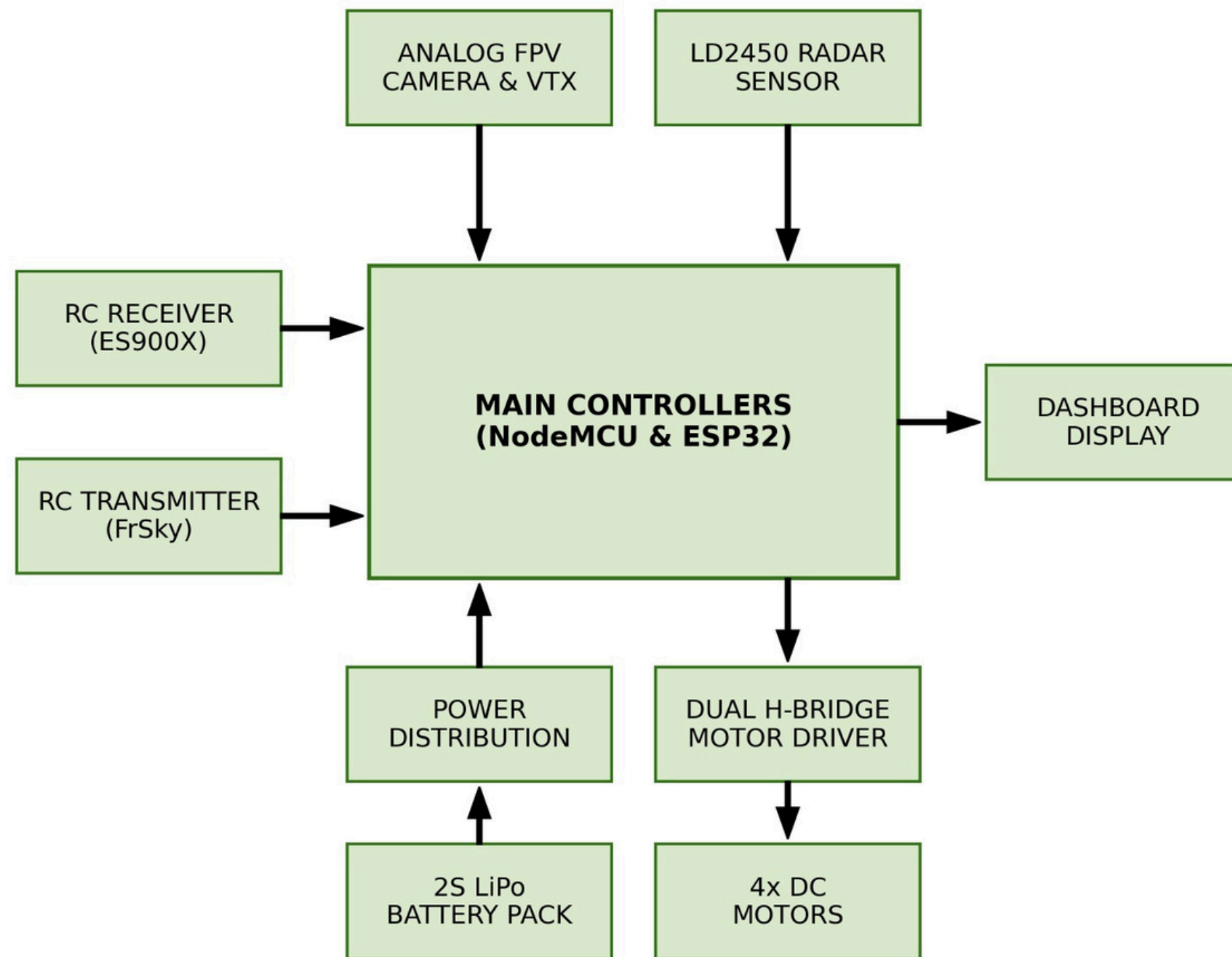


Fig 3 : System Block diagram

Methodology

- Hardware Design

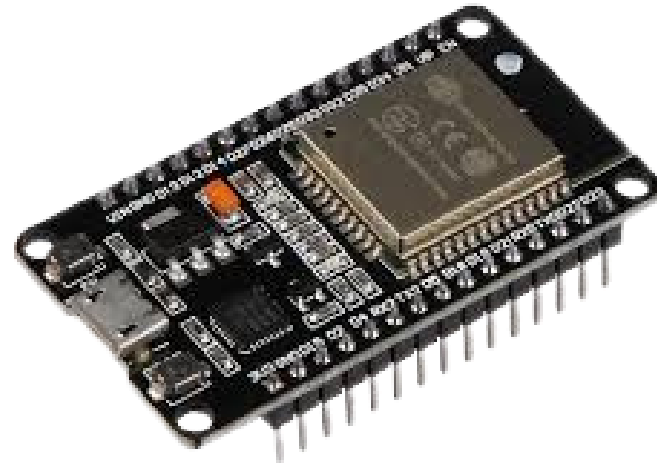


Fig 4 : ESP 32[10]

- Central processing unit
- Executes analytics algorithms
- Hosts embedded web dashboard
- Provides WiFi connectivity

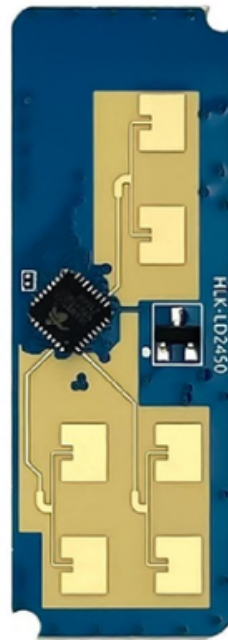


Fig 5 : HLK-LD2450[10]

- 24GHz FMCW radar sensor
- Tracks up to 3 targets simultaneously
- Provides position and velocity data
- Supports zone-based detection

Methodology

- Hardware Design



Fig 4 : CrowPanel Advance 7" ESP32-S3 display[10]

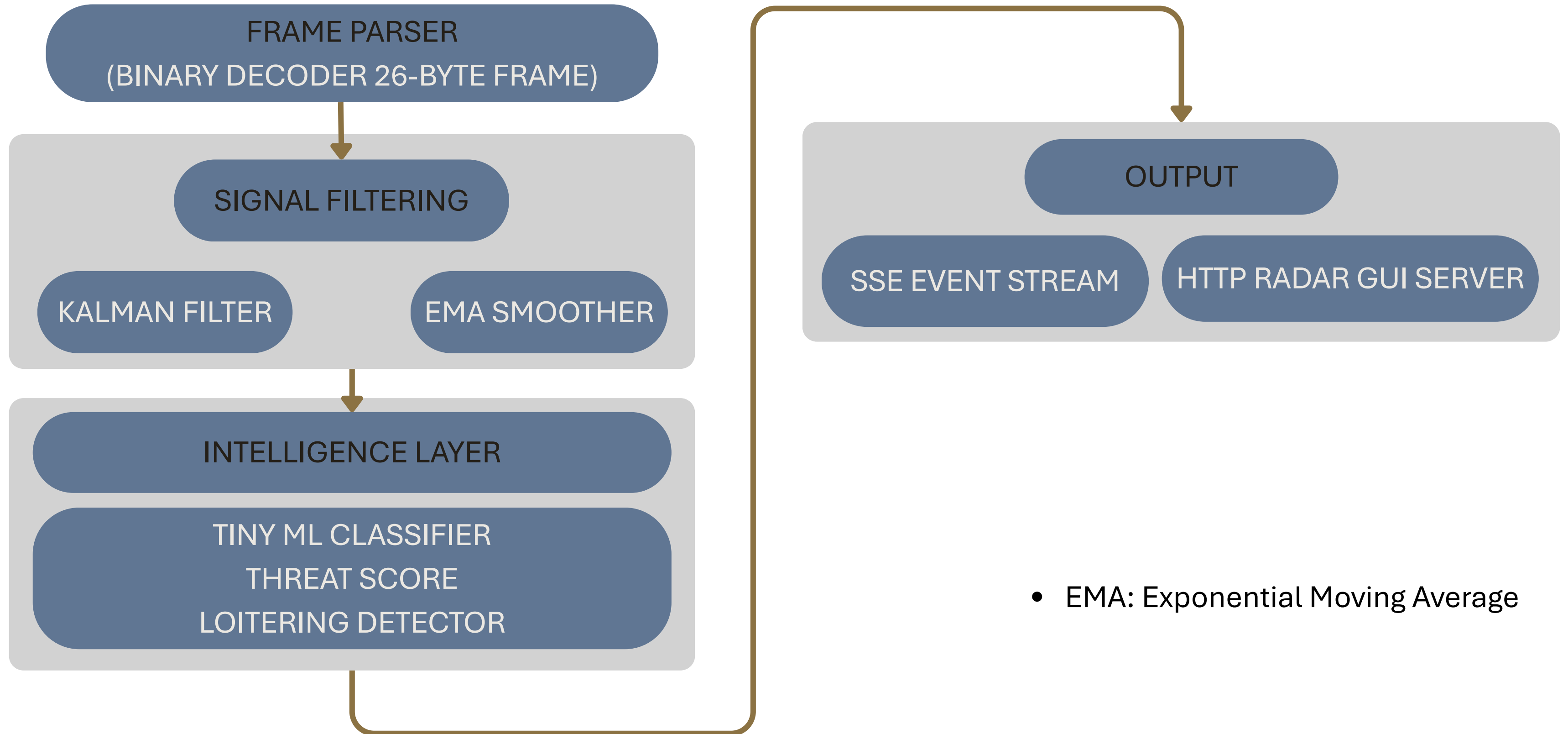
- Provides real-time visual interface
- Shows target positions, motion states, and alerts
- Supports user interaction and system status monitoring



Fig 5 : FrSky Taranis Q X7S transmitter[10]

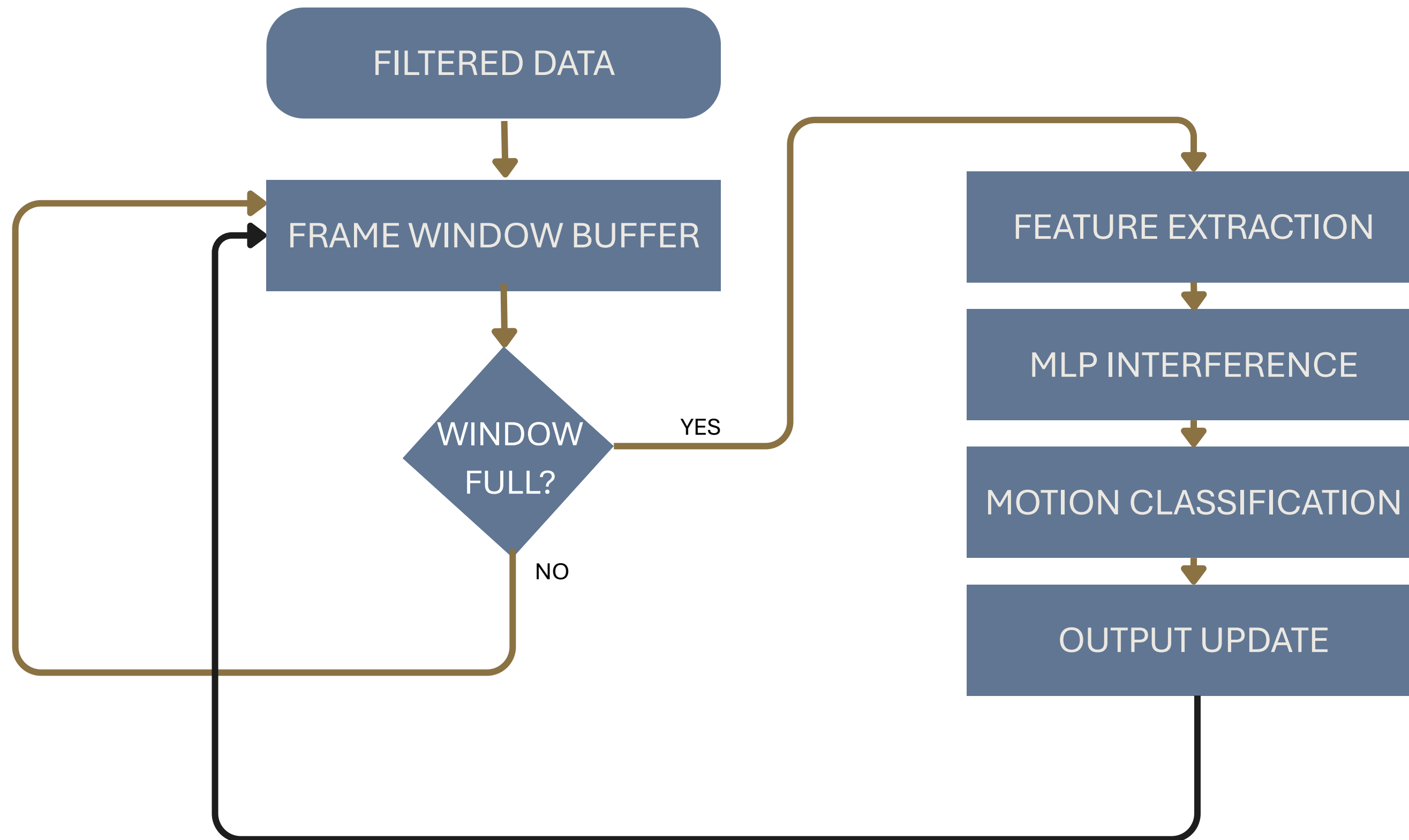
- Enables remote teleoperation of the rover
- Provides precise control of motion and operating modes
- Supports responsive, low-latency surveillance operation

ESP32 FIRMWARE ARCHITECTURE



- EMA: Exponential Moving Average

TINYML MOTION CLASSIFICATION PIPELINE



CIRCUIT DIAGRAM

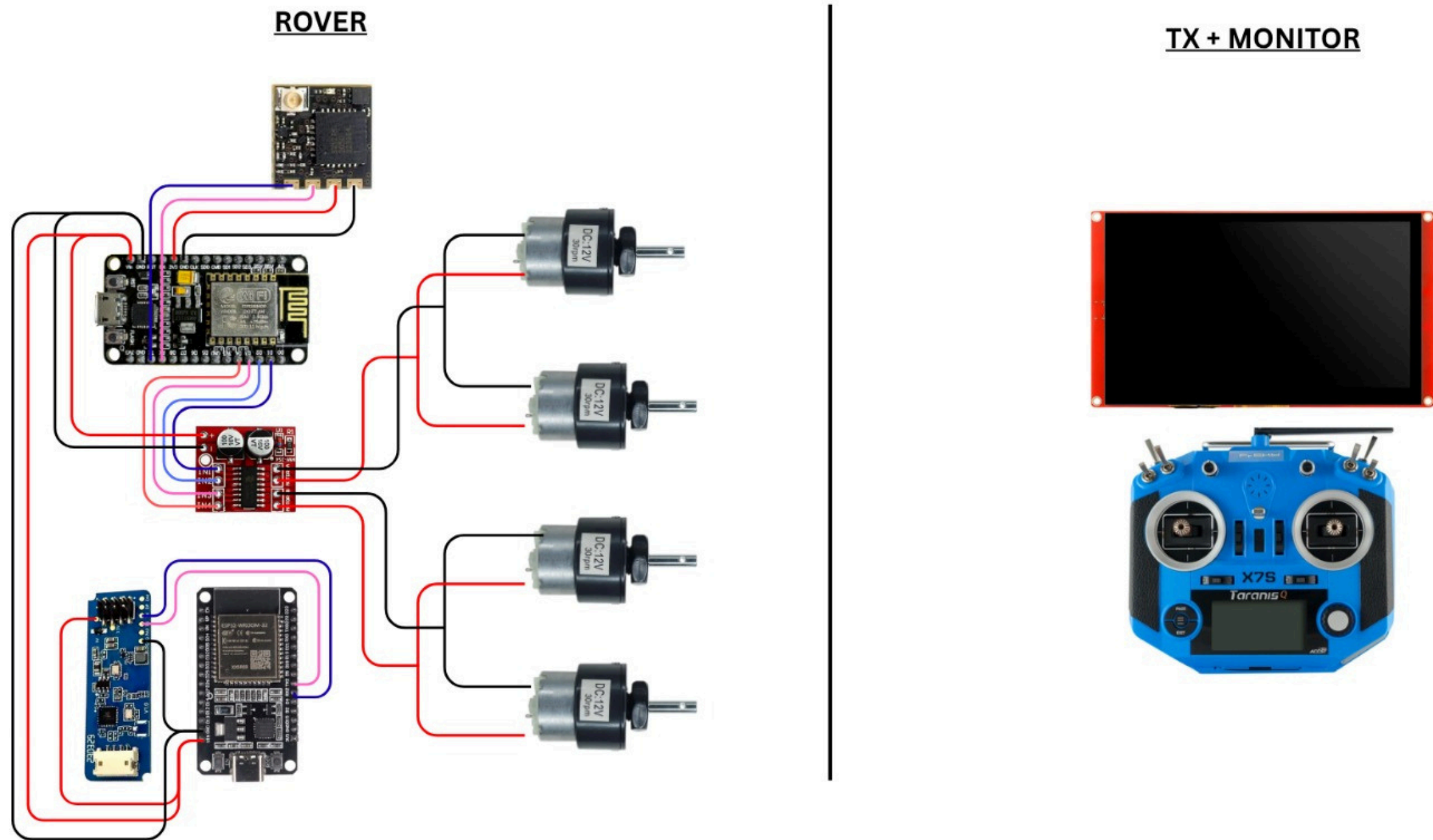


Fig 7 : Circuit Diagram

EXPERIMENTAL SETUP

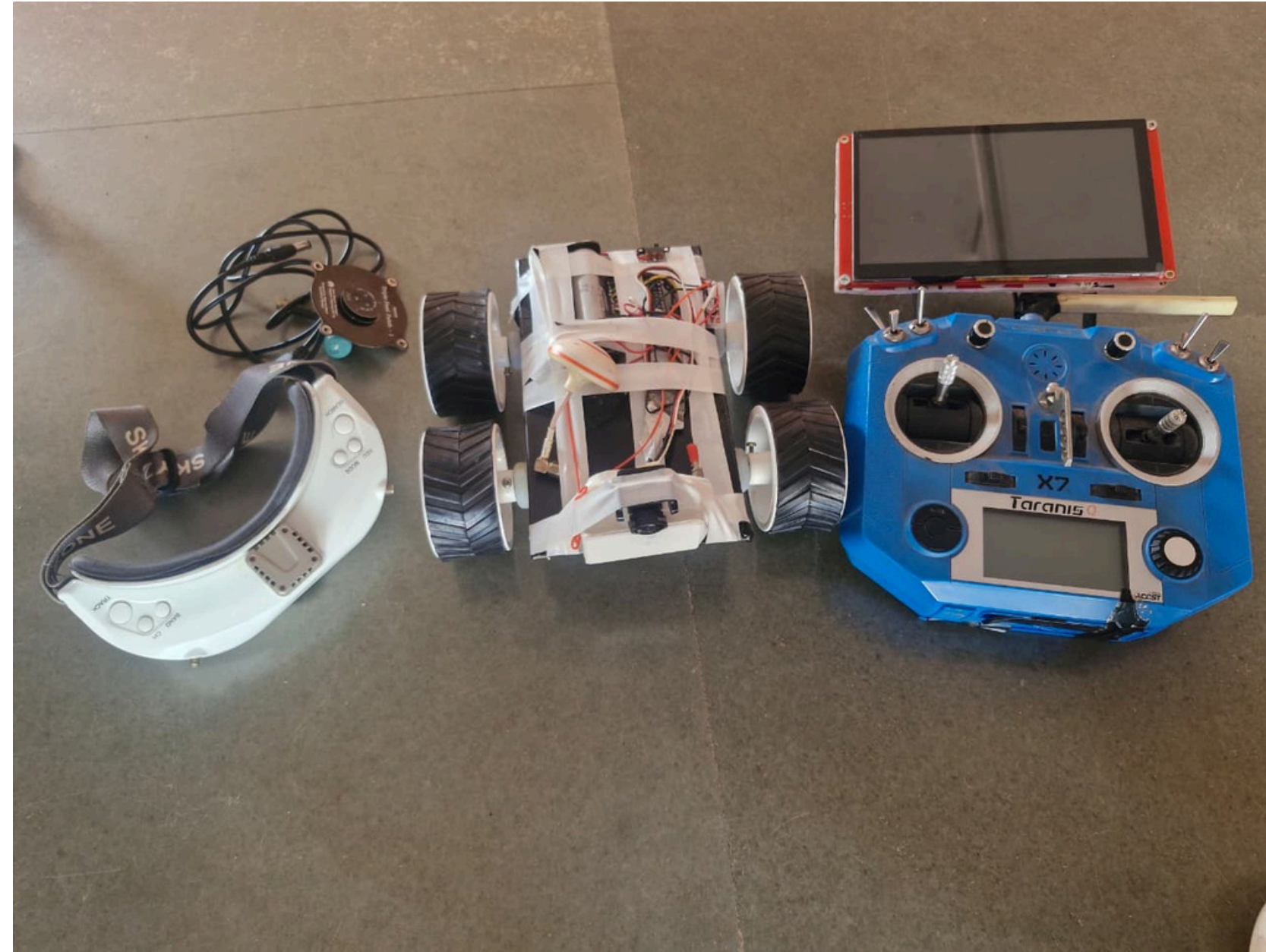


Fig 8 : Experimental Setup

RESULTS AND ANALYSIS

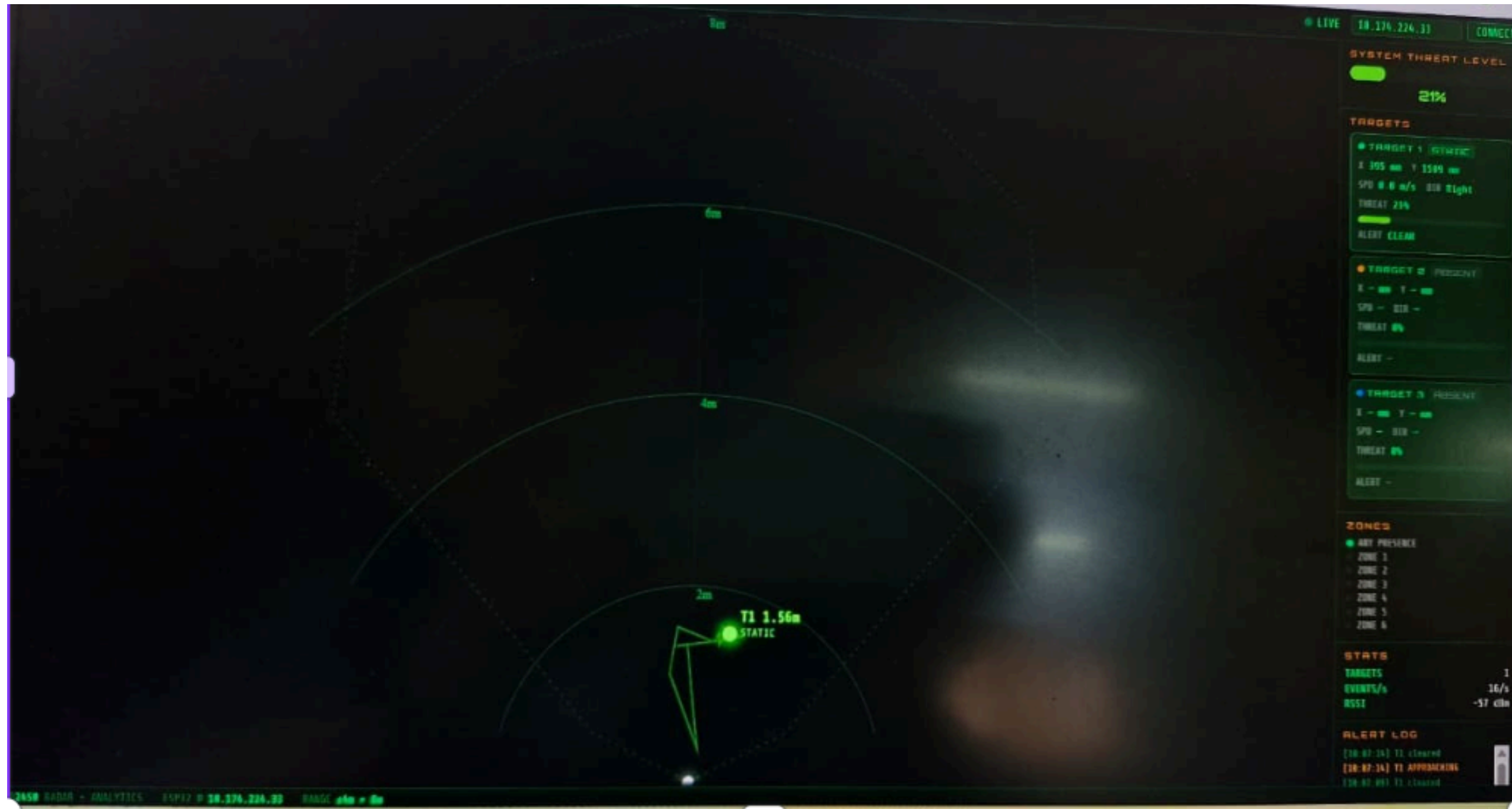


Fig 9 : Radar GUI

RESULTS AND ANALYSIS

Kalman Filter Noise Reduction

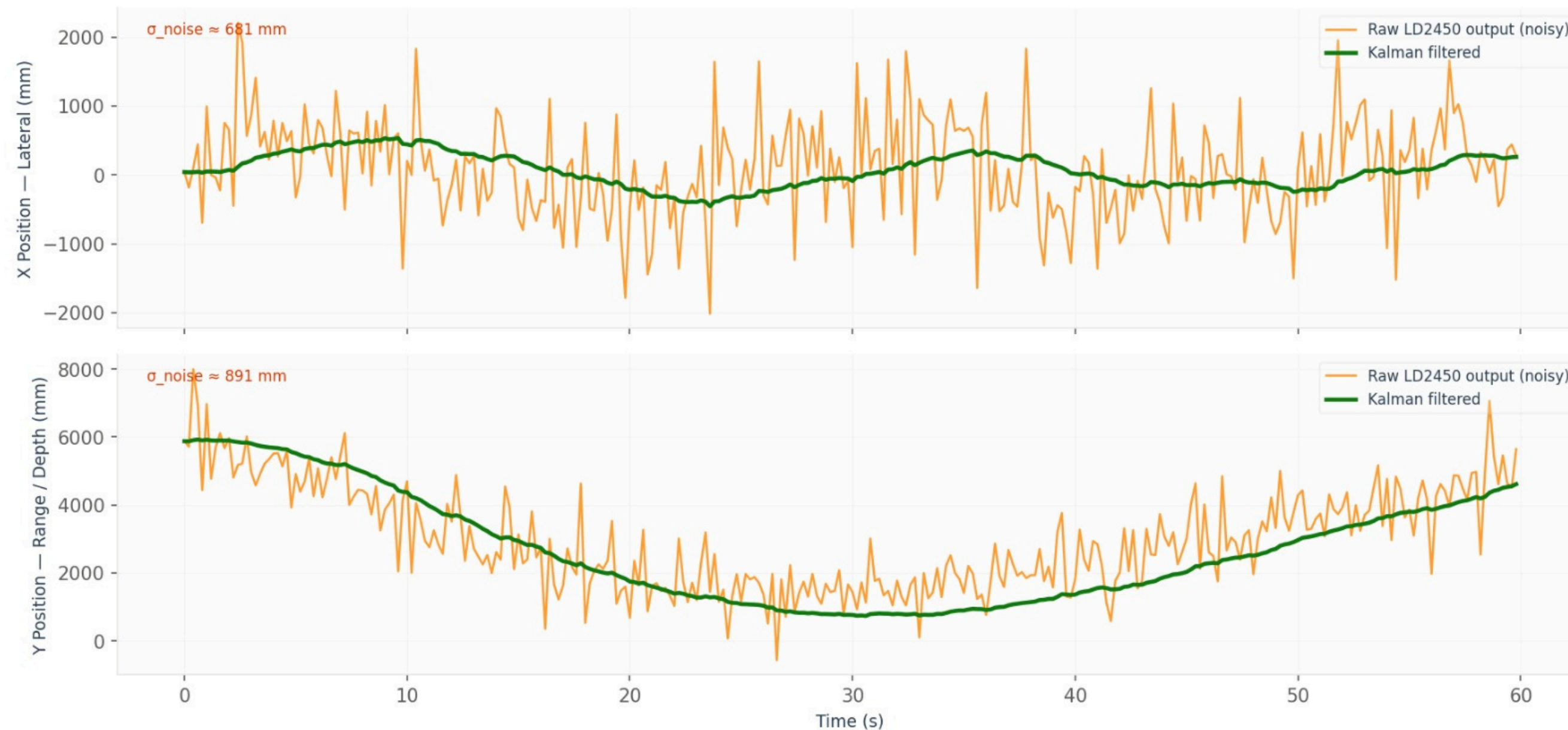


Fig 10 : Kalman Filter Noise Reduction on LD2450 Position Data

RESULTS AND ANALYSIS

Kalman Filter Noise Reduction

- This represents the root-mean-square (RMS) of the random jitter in the raw LD2450 radar data.
- The LD2450 is a high-sensitivity radar and can produce "spiky" raw position data. The Kalman Filter uses a mathematical model of human motion to strip away this jitter.
- A higher σ_{noise} (e.g., > 300mm) indicates a very noisy environment or a fast-moving target causing sensor fluctuations. The green line in the graph shows the "clean" path after this noise is suppressed.

RESULTS AND ANALYSIS

Real Time Threat Score Timeline



Fig 11 : Real Time Threat Score Timeline with Alert State Transitions

RESULTS AND ANALYSIS

Real Time Threat Score Definition

The Threat Score is a real-time composite metric (0–100) used to assess the potential risk posed by a detected target. It is calculated using three primary physical components:

Component	Weight	Calculation Logic
Speed (t_spd)	35%	$\min(35.0, \text{speed_m_s} * 14.0)$. Higher speed linearly increases the threat up to 2.5 m/s.
Approach (t_app)	40%	Based on the Dot Product between the target's movement vector and the vector pointing toward the sensor. Only targets directly approaching the sensor receive this score.
Proximity (t_prox)	25%	$\max(0.0, 25.0 * (1 - \text{distance_mm} / 7000.0))$. Targets closer than 7 meters receive a proximity score that peaks at 0mm.

RESULTS AND ANALYSIS

Real Time Threat Score Definition

Alert Thresholds:

- CLEAR: 0–45
- APPROACHING: 45–72
- THREAT: 72–100
- LOITERING: Triggered if a target stays within a 600mm radius for more than 12 seconds.

2D TARGET POSITION TRACE

Scatter plot of raw vs Kalman-filtered positions in mm. Color-coded by threat score; blue arrows indicate predicted movement direction.

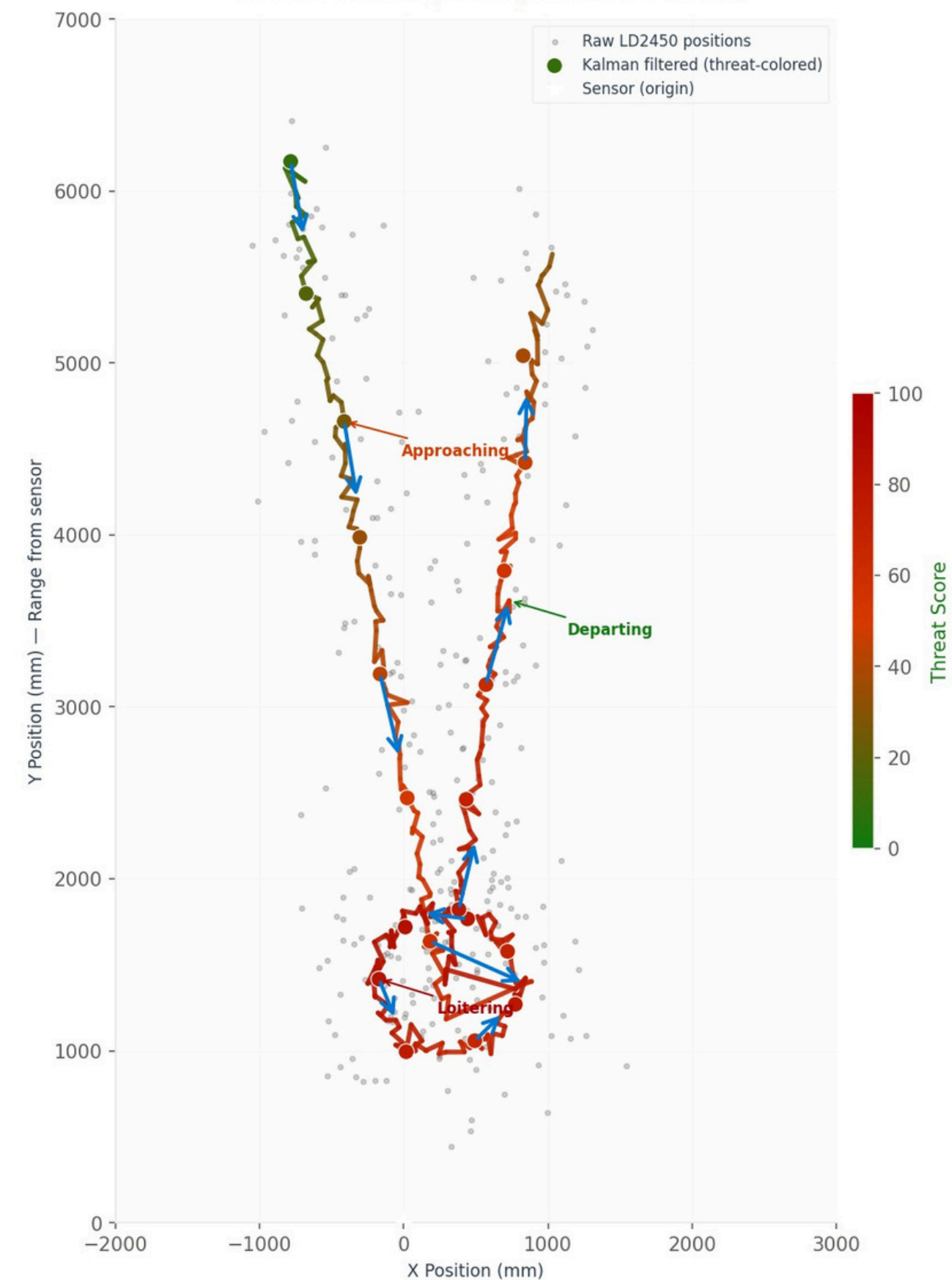


Fig 12 : 2D Target Position Trace(Kalman Filtered+Prediction Vectors)

DATA SAMPLE COLLECTION

- The system's TinyML model was trained and validated on a dataset specifically collected for five distinct motion profiles.
- Total Collected Samples: 5,819 frames (@ 5Hz)

CLASSIFICATION	SAMPLE COUNT
STATIC	1348 Samples
CREEPING	462 Samples
WALKING	1345 Samples
RUNNING	10048 Samples
PACING	1616 Samples

CONFUSION MATRIX HEATMAP

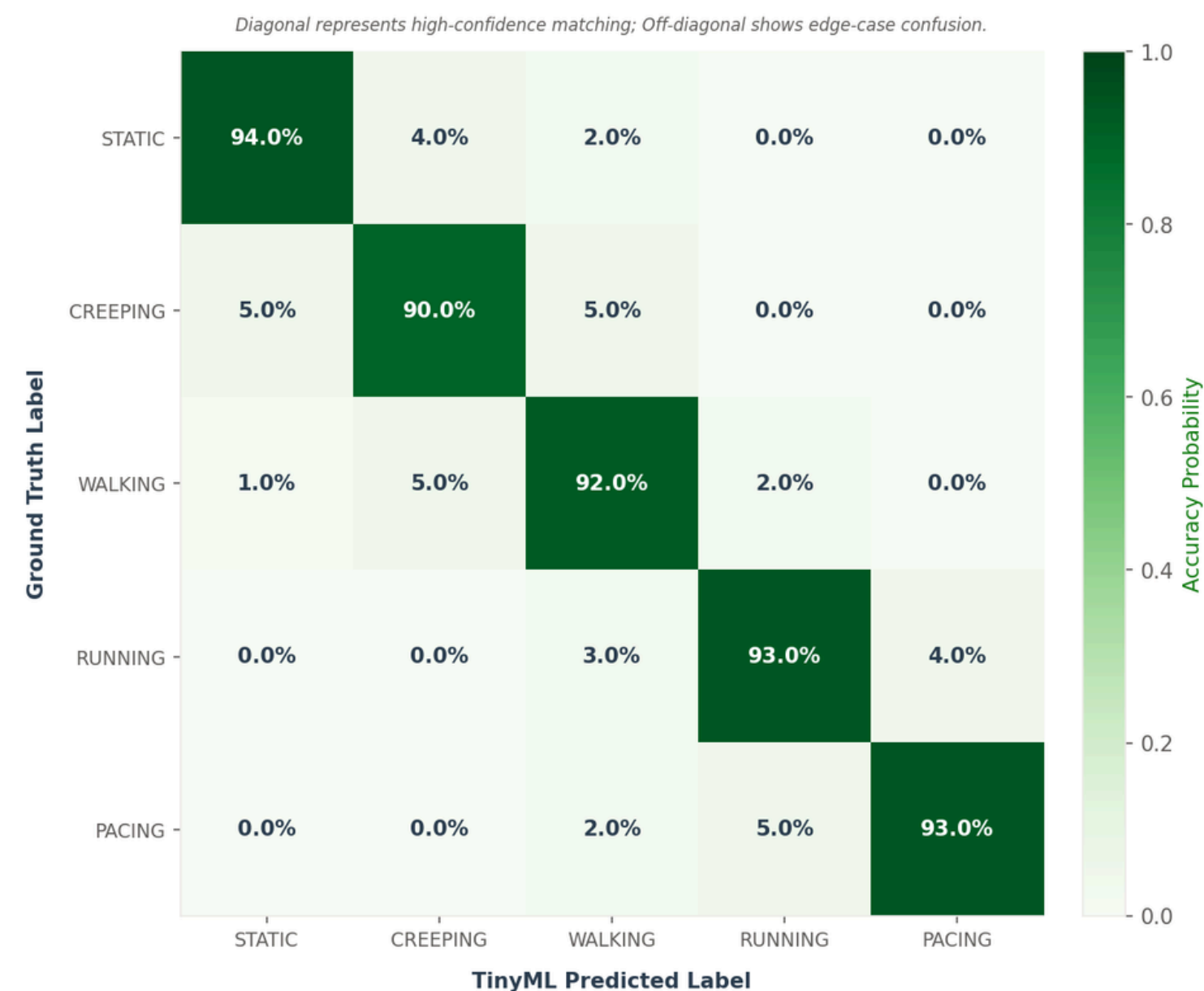


Fig 13 : Confusion Matrix Heatmap

MOTION CLASSIFICATION DISTRIBUTION

- Average Classification accuracy is 90.08%

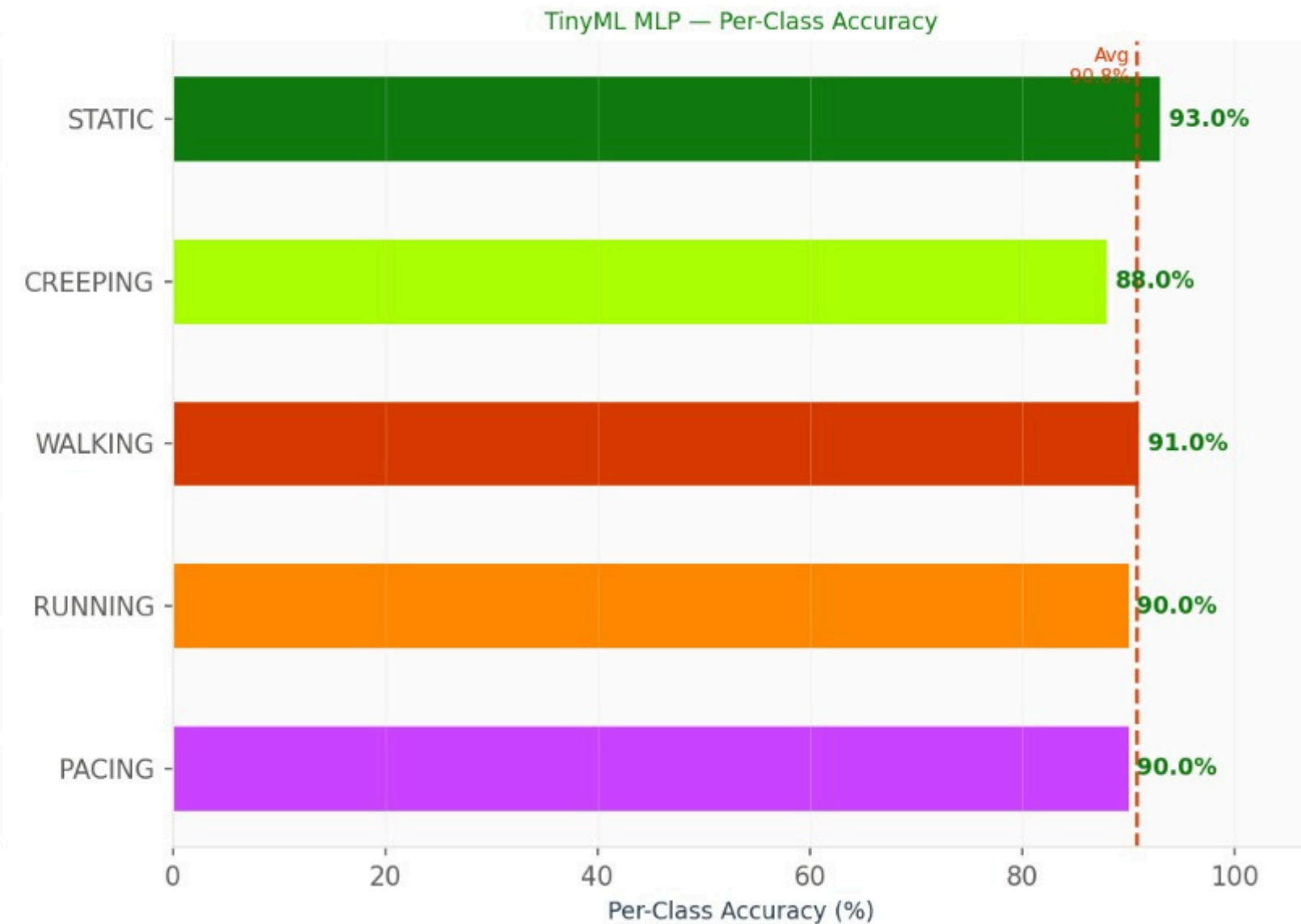
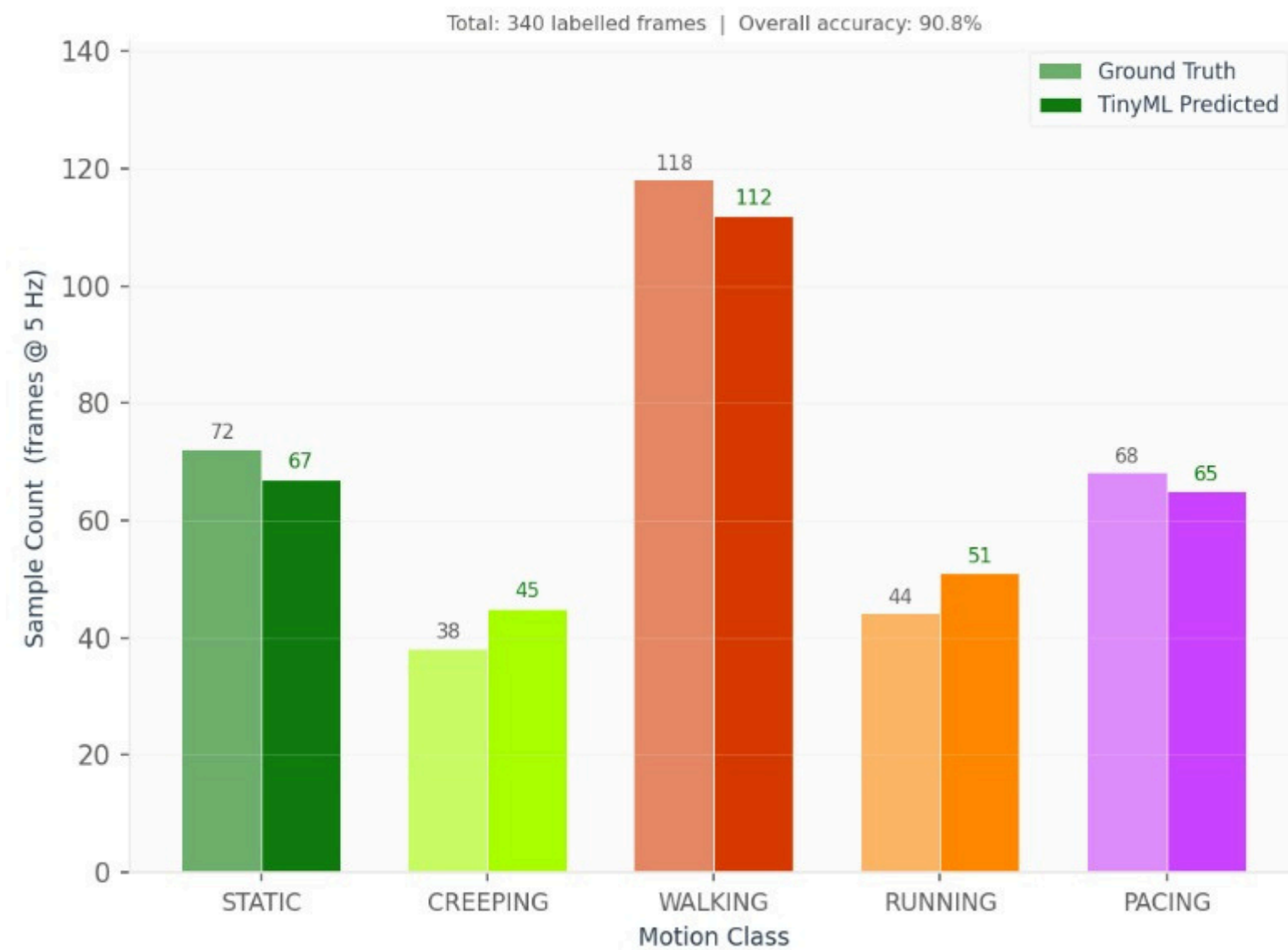


Fig 14 : Target Classification Results

SYSTEM COMPARISON CHART

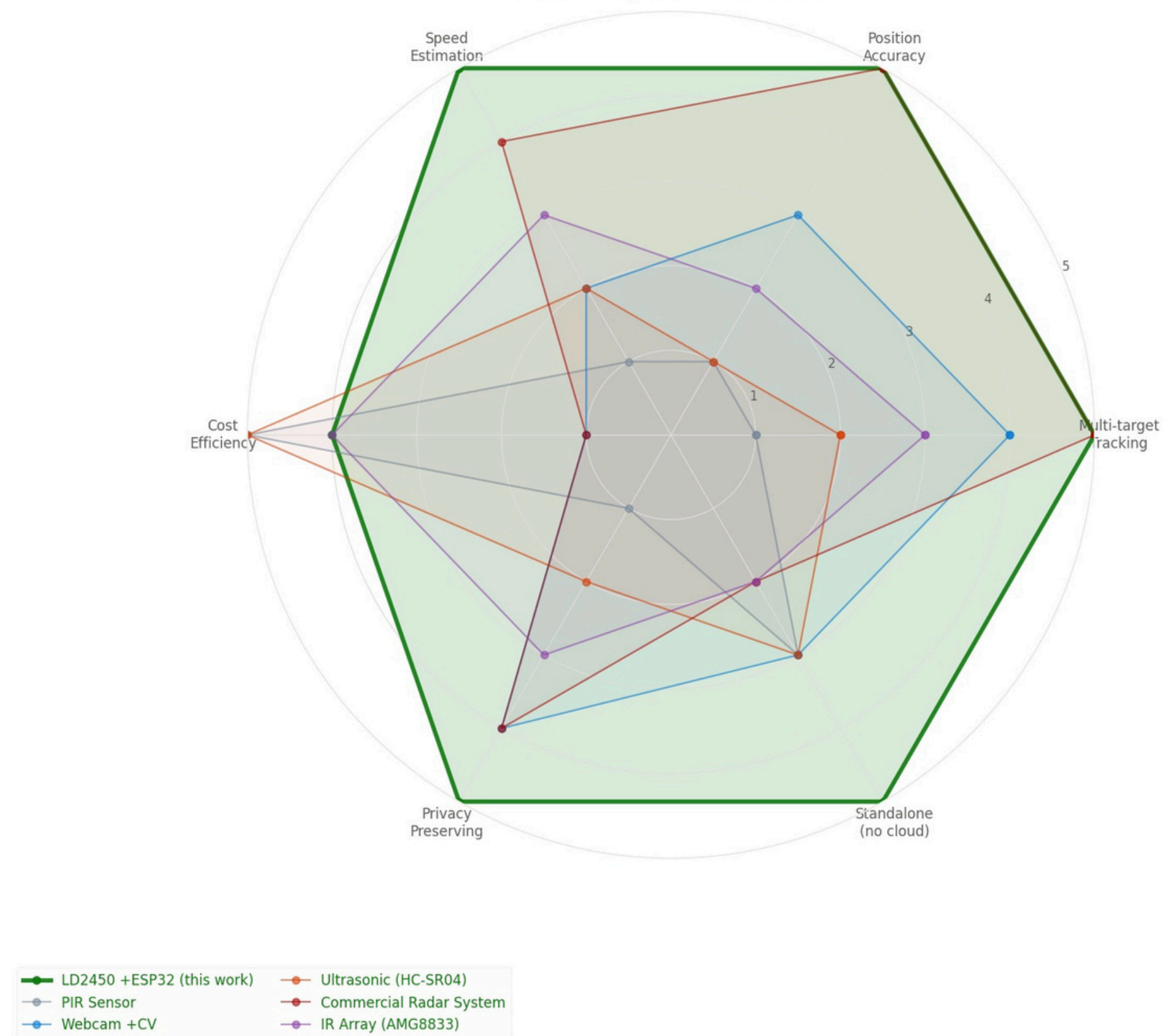


Fig 15 :LD2450+ESP32 Vs Alternatives

Project TimeLine (Gantt Chart)

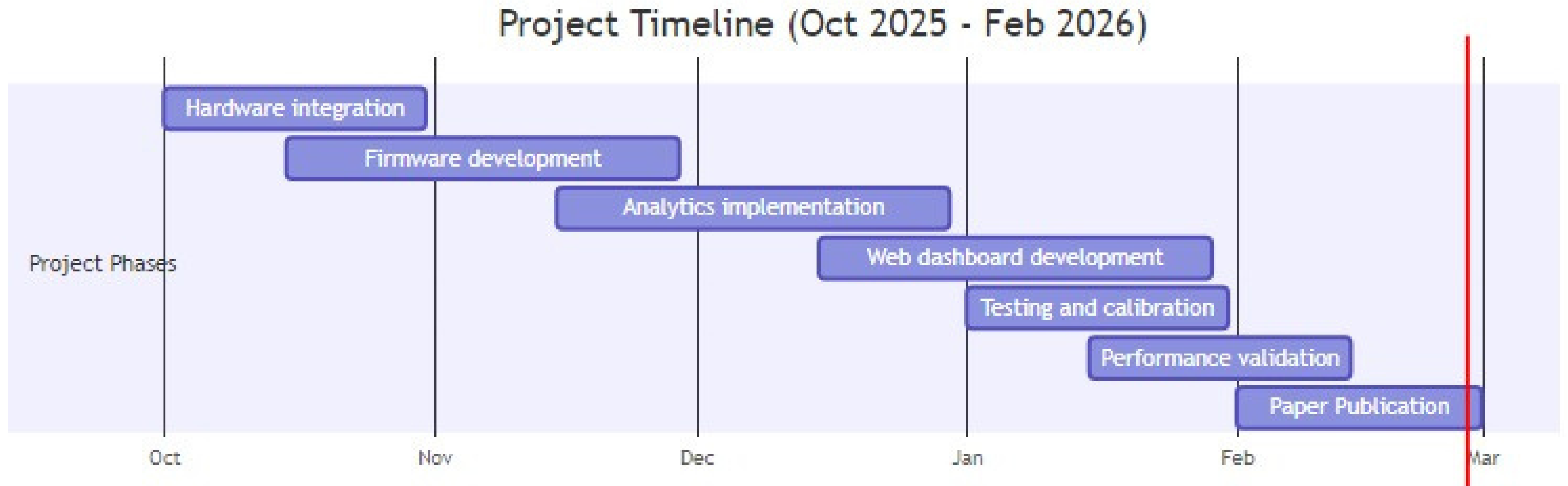


Fig 16 : Gantt chart

Project Budget

SlNo	Product Name	Quantity	Price
1	HLK-LD2450 Radar	1	Rs. 1640
2	ESP32	1	Rs. 360
3	CrowPanel 7" ESP32-S3 Display	1	Rs. 5000
-	TOTAL	2	Rs. 7000

Individual Contribution

1. Project Prototype and ML Training - Arjun M
2. Conference Paper Content and ML , Project Report, Presentation - Kripa Krishna
3. Hardware and Firmware Testing - Blesson Binu

Reference

- [1] Ivanovs, Arturs, et al. "Multisensor low-cost system for real time human detection and remote respiration monitoring." 2019 Third IEEE international conference on robotic computing (IRC). *IEEE*, 2019.
- [2] Wang, Weihang, et al. "A high-computational efficiency human detection and flow estimation method based on tof measurements." *Sensors* 19.3 (2019): 729.
- [3] Liu, Chang, and Tamás Szirányi. "Real-time human detection and gesture recognition for on-board UAV rescue." *Sensors* 21.6 (2021): 2180.
- [4] Mantau, Aprinaldi Jasa, et al. "A human-detection method based on YOLOv5 and transfer learning using thermal image data from UAV perspective for surveillance system." *Drones* 6.10 (2022): 290.
- [5] Tsai, Pei-Fen, Chia-Hung Liao, and Shyan-Ming Yuan. "Using deep learning with thermal imaging for human detection in heavy smoke scenarios." *Sensors* 22.14 (2022): 5351.
- [6] Gu, Zhanzhong, et al. "Millimeter wave radar-based human activity recognition for healthcare monitoring robot." *arXiv preprint arXiv:2405.01882* (2024).
- [7] El Hail, Reda, et al. "Radar-Based Human Activity Recognition: A Study on Cross-Environment Robustness." *Electronics* 14.5 (2025): 875.
- [8] Gao, Weicheng. "Through-the-Wall Radar Human Activity Recognition WITHOUT Using Neural Networks." *arXiv preprint arXiv:2506.05169* (2025).

Bibiliography

- [9] Google. (2025, Dec. 30). "Visual Indicators of High Stress Vs Neutral Intent." [AI-Generated Image]. Generated by Gemini.
- [10] "Comparative Study of Microcontroller: ARDUINO UNO, RASPBERRY PI 4, ESP 32" (2024). International Journal for Research in Applied Science and Engineering Technology (IJRASET).

LIST OF PUBLICATIONS

- Presented paper “Enhanced Target Classification in FMCW Radar via Deeper Spatio – Temporal CNN-LSTM Architecture” in International Conference on Fostering Innovation in Intelligent Systems and Next Generation Network 2026 (ICFISN 2026)
- Submitted paper “Deep Learning-Based Target Classification Using Compact FMCW Radar MTI Representations” in International Conference on Smart Communication and Sustainable Technologies 2026 (ICSCST 2026)